Kintali's counterexample to fixed-order prefix balancing

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Abstract

The strong Komlós question asks whether one signing can keep every prefix of any fixed ordered family of Euclidean-unit vectors bounded in the maximum norm by an absolute constant. We give a self-contained exposition of Kintali's counterexample. His finite recursive family has dimension $d_k = 2^{2^k-1}$ and prefix discrepancy at least $2^{k/2-1} = \frac{1}{2}\sqrt{1+\log_2 d_k}$ for every signing. The proof freezes a large mean in each of several coordinate pairs and then forces a large difference in one of those pairs. All witness choices are represented in a fixed ordered word, so the argument applies to signs chosen with knowledge of the entire input. The construction and amplification argument are due to Kintali.

1 The question and its source

For a matrix $A = (v_1, \dots, v_T)$ with ordered columns in $\mathbb{R}^d$, define

$$\operatorname{pdisc}(A) = \min_{\varepsilon \in \{-1,1\}^T} \max_{1 \le t \le T} \left\| \sum_{i=1}^{t} \varepsilon_i v_i \right\|_\infty, \qquad \|x\|_\infty = \max_{1 \le r \le d} |x_r|.$$

The order is fixed, and the same signs must work for all prefixes. They may depend on the entire input. The strong Komlós conjecture asks whether $\operatorname{pdisc}(A)$ has an absolute upper bound whenever $\|v_i\|_2 \le 1$. Bansal et al. [1, Open Problem 6.1] state this question, which is the target of AIM problem 325 [4]. It differs from the terminal-sum problem: a small value of $\|\sum_i \varepsilon_i v_i\|_\infty$ allows large intermediate excursions. Karingula and Lovett's terminal-sum result and their Conjecture 1.6 explicitly retain this distinction [2].

Kintali's preprint [3, Lemma 2.2 and Theorem 3.1] gives the following negative answer. This note explains its construction and proof; it makes no claim of a new construction or mathematical priority.

Theorem 1 (Kintali)**.** *For every integer $k \ge 0$, there is a finite fixed ordered matrix $A_k \in \mathbb{R}^{d_k \times n_k}$, where $d_k = 2^{2^k-1}$ and $n_k \ge 1$, whose columns have Euclidean norm one, such that every $\varepsilon \in \{-1,1\}^{n_k}$ satisfies*

$$\max_{1 \le t \le n_k} \left\| \sum_{i=1}^{t} \varepsilon_i (A_k)_i \right\|_\infty \ge 2^{k/2-1} = \frac{1}{2}\sqrt{1+\log_2 d_k}. \tag{1}$$

Consequently there is no absolute constant in the strong Komlós question.

Here $(A_k)_i$ is column i, and n_k is its finite number of columns. The right side grows without bound. The length of the construction is large; neither an efficient algorithm nor an optimal lower bound is asserted.

2 The construction and amplification

Copies of a matrix can begin after earlier copies have moved the current sum. We therefore use a statement that allows an arbitrary initial offset. Put $E = \{c \in \mathbb{R} : |c| \geq 1\}$. For an ordered $d \times n$ matrix $M = (M_1, \ldots, M_n)$ and $a \geq 0$, write $R(M, a)$ for

$$\forall c \in E^n \ \forall u \in \mathbb{R}^d : \quad \max_{0 \leq t \leq n} \left\| u + \sum_{j=1}^{t} c_j M_j \right\|_\infty \geq a. \tag{2}$$

The prefix with $t = 0$ is included in this auxiliary property.

Assume $d, n \geq 1$. In dimension $2d^2$, index coordinates by (ℓ, r, σ) for $\ell, r \in [d] = \{1, \ldots, d\}$ and $\sigma \in \{+, -\}$, and let $e_{\ell,r,\sigma}$ be the corresponding standard vector. For each source column j, define

$$O_{\ell,j} = \sum_{r=1}^{d} m_{rj}(e_{\ell,r,+} + e_{\ell,r,-}), \tag{3}$$

$$D_{\rho,j} = \sum_{\ell=1}^{d} m_{\ell j}(e_{\ell,r_\ell,+} - e_{\ell,r_\ell,-}), \qquad \rho = (r_1, \ldots, r_d) \in [d]^d. \tag{4}$$

The source row in the second formula is ℓ; r_ℓ chooses only the physical coordinate pair. Different depths ℓ use disjoint pairs. An original column will change the mean of a pair, while a detector column will change its difference.

Let Q concatenate $(D_{\rho,1}, \ldots, D_{\rho,n})$ for every $\rho \in [d]^d$ in lexicographic order. Define words backwards by

$$W_{d+1} = Q, \qquad W_\ell = (W_{\ell+1}, O_{\ell,1}, W_{\ell+1}, \ldots, O_{\ell,n}, W_{\ell+1}) \quad (\ell = d, \ldots, 1). \tag{5}$$

The matrix $F(M)$ has ordered column word W_1. Every physical occurrence has its own coefficient, even when two occurrences are equal as vectors. The entire word is specified before these coefficients are chosen.

Each output column corresponding to M_j has squared norm $2\|M_j\|_2^2$. This follows by counting its two copies of each source entry in distinct coordinates in (3) and (4). In particular, s nonzero entries in $\{-1, 1\}$ per source column become exactly $2s$ such entries per output column. The output dimension and length are

$$d' = 2d^2, \qquad n' = (1 + nd^d)(n+1)^d - 1. \tag{6}$$

Indeed $|Q| = nd^d$ and $|W_\ell| + 1 = (n+1)(|W_{\ell+1}| + 1)$. Thus the construction is finite for every finite source matrix.

Lemma 2 (Kintali)**.** *If $R(M, a)$ holds, then $R(F(M), 2a)$ holds.*

Proof. Fix arbitrary coefficients in E on all n' physical occurrences and an arbitrary initial vector $w \in \mathbb{R}^{2d^2}$. Let $p(t)$ be the resulting path, including $p(0) = w$. For each coordinate pair set

$$h_{\ell,r}(t) = \frac{p_{\ell,r,+}(t) + p_{\ell,r,-}(t)}{2}, \qquad z_{\ell,r}(t) = \frac{p_{\ell,r,+}(t) - p_{\ell,r,-}(t)}{2}.$$

Thus $p_{\ell,r,\pm} = h_{\ell,r} \pm z_{\ell,r}$. An occurrence of $O_{\ell,j}$ changes only the depth-ℓ mean vector, by its coefficient times M_j. No original changes any z. A detector $D_{\rho,j}$ changes z_{ℓ,r_ℓ} by its coefficient times $m_{\ell j}$ for every ℓ, and changes no mean.

In an occurrence of W_ℓ, let h^0 be the depth-ℓ mean vector at its start and $\alpha_j \in E$ the coefficients of its n separating original columns. At the starts of its $n+1$ child occurrences of $W_{\ell+1}$ the mean vector has values

$$h^0 + \sum_{j=1}^{s} \alpha_j M_j, \qquad 0 \le s \le n.$$

Property (2) selects a child and a row r_ℓ with $|h_{\ell,r_\ell}| \ge a$. That mean stays fixed throughout the chosen child, since it contains only deeper originals and detectors. Starting at W_1, repeat this selection inside each chosen child. After d selections there is one occurrence of Q and rows $r_1, \dots, r_d$ such that all means h_{ℓ,r_ℓ} are constant there and have absolute value at least a. Previously selected means remain fixed on every descendant interval.

This occurrence of Q contains the detector word for $\rho = (r_1, \dots, r_d)$. If its occurrence coefficients are $\beta_1, \dots, \beta_n$, the vector of selected differences follows

$$(z_{1,r_1}, \dots, z_{d,r_d})_{\text{start}} + \sum_{j=1}^{s} \beta_j M_j, \qquad 0 \le s \le n.$$

Its initial value may reflect earlier detectors. Property (2) permits precisely this arbitrary offset, and gives a time and depth ℓ with $|z_{\ell,r_\ell}| \ge a$. At the same time $|h_{\ell,r_\ell}| \ge a$. For real h, z,

$$\max\{|h+z|, |h-z|\} = |h| + |z|,$$

as follows by squaring both sides. One physical coordinate therefore has magnitude at least $2a$ at a global prefix time. The initial vector and all occurrence coefficients were arbitrary. Choosing the witness interval and tuple after fixing them changes neither the input word nor its coefficients. This proves the lemma. □

3 Iteration and unit normalization

Proof of Theorem 1. Set $M_0 = [1]$ and $M_{k+1} = F(M_k)$. For $u \in \mathbb{R}$ and $|c| \ge 1$,

$$|c| \le |u| + |u + c|,$$

so $R(M_0, 1/2)$ holds. Lemma 2 gives $R(M_k, 2^{k-1})$ for every integer $k \ge 0$. Each M_k is an integer matrix with exactly s_k nonzero entries in $\{-1, 1\}$ per column. The norm count and (6) give

$$s_0 = d_0 = n_0 = 1, \qquad s_{k+1} = 2s_k, \qquad d_{k+1} = 2d_k^2,$$

and the finite positive length obeys

$$n_{k+1} = (1 + n_k d_k^{d_k})(n_k + 1)^{d_k} - 1.$$

Induction yields $s_k = 2^k$ and $d_k = 2^{2^k - 1}$. Define $A_k = M_k / \sqrt{s_k}$. Every column then has squared Euclidean norm $s_k / s_k = 1$. For any signing, apply $R(M_k, 2^{k-1})$ with offset zero and divide by $\sqrt{s_k}$. The result is (1), since $2^{k-1}/\sqrt{2^k} = 2^{k/2-1}$ and $1 + \log_2 d_k = 2^k$. The zero prefix has norm zero, so it cannot be the witness to this positive bound.

For any proposed $C > 0$, choose a positive integer $m > 2C$ and take $k = 2m$. Since $2^m \ge m + 1 > 2C$, the bound $2^{k/2-1}$ exceeds C. The fixed finite family A_k then defeats every signing. Every column is retained, the order is unchanged, and the signs may use the entire family. □

AI-use statement. OpenAI Codex assisted with source retrieval, reconstruction and checking of the argument, and preparation of this exposition. The construction and proof mechanism are credited to Shiva Kintali. The checking performed for this note was AI-assisted; no independent human review is claimed.

References

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[2] S. R. Karingula and S. Lovett, *An elementary proof of the Komlós conjecture*, ECCC TR26-188, 2026. arXiv:2609.20979v1, Theorem 1.2 and Conjecture 1.6.

[3] S. Kintali, *On the Fixed-Order Strong Komlós Conjecture*, ECCC TR26-242, 2026, Sections 2–3.1, Lemma 2.2 and Theorem 3.1, pp. 2–5. The PDF title page is dated September 15; the ECCC report record is dated September 26; its publication field gives October 9.

[4] *AIM—Open Applied Problems*, problem 325, fixed-order prefix discrepancy statement, revision `c9929805cece044705abaaebe1cebddac1ba11a6`, accessed October 12, 2026.